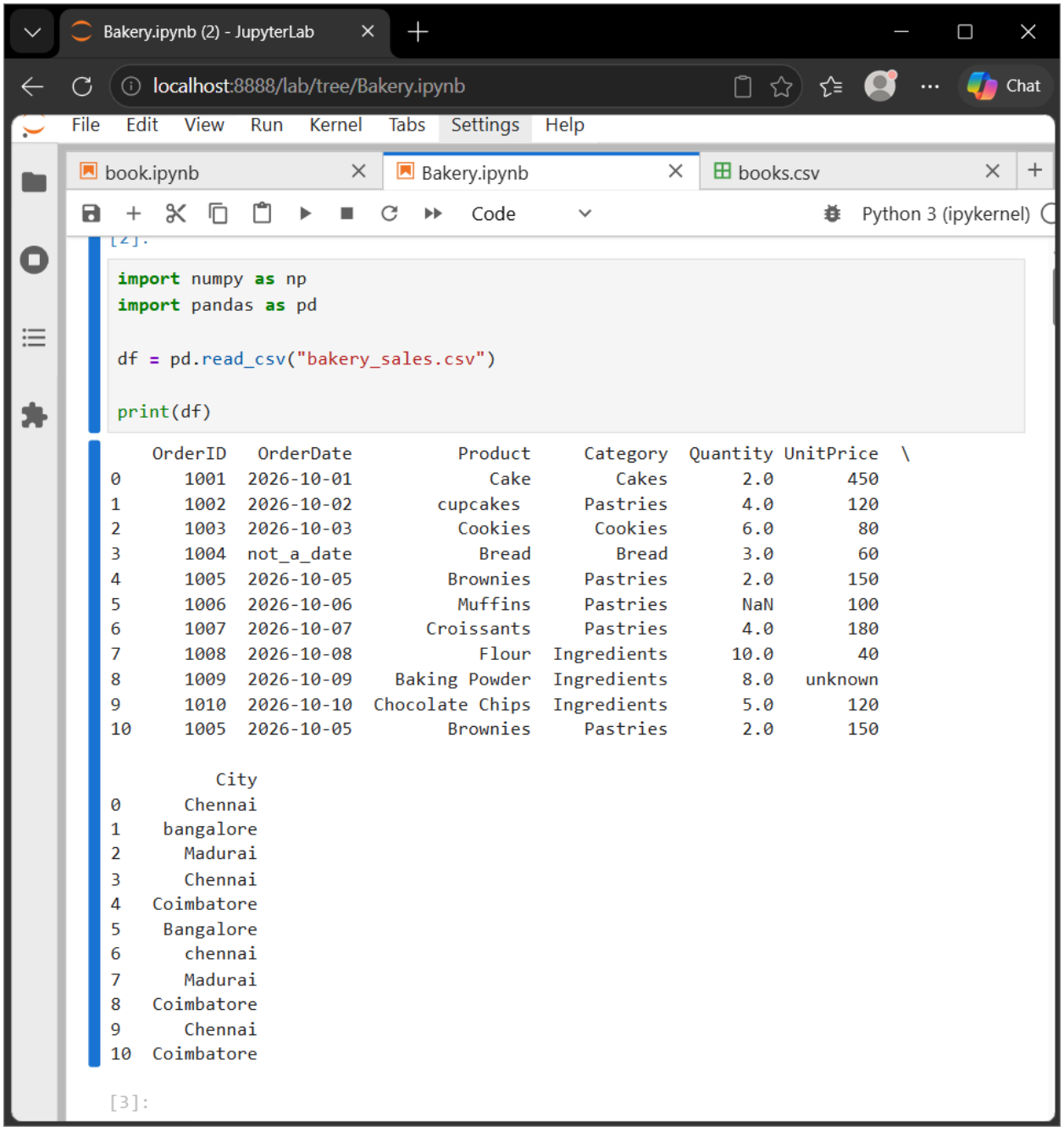
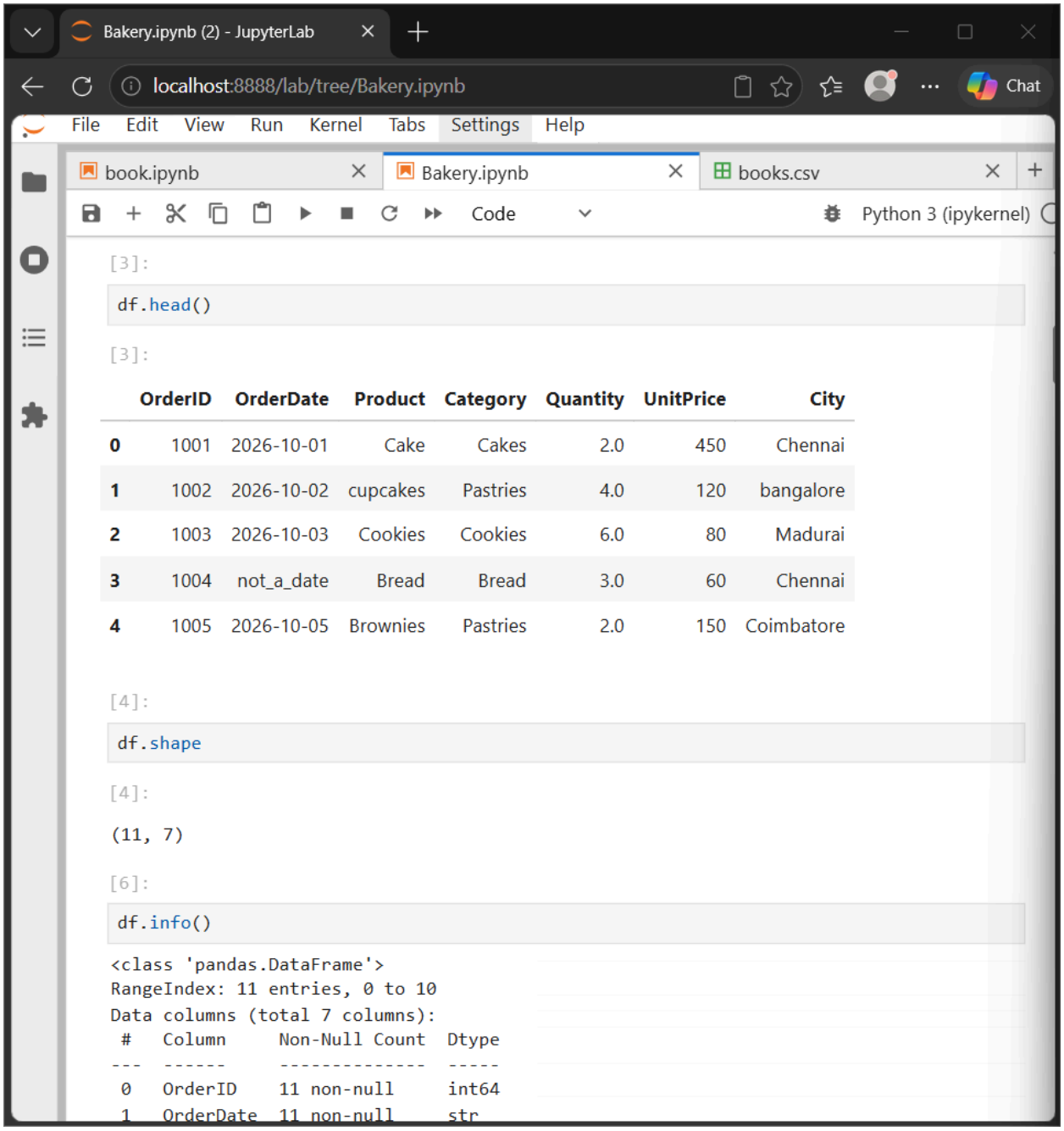
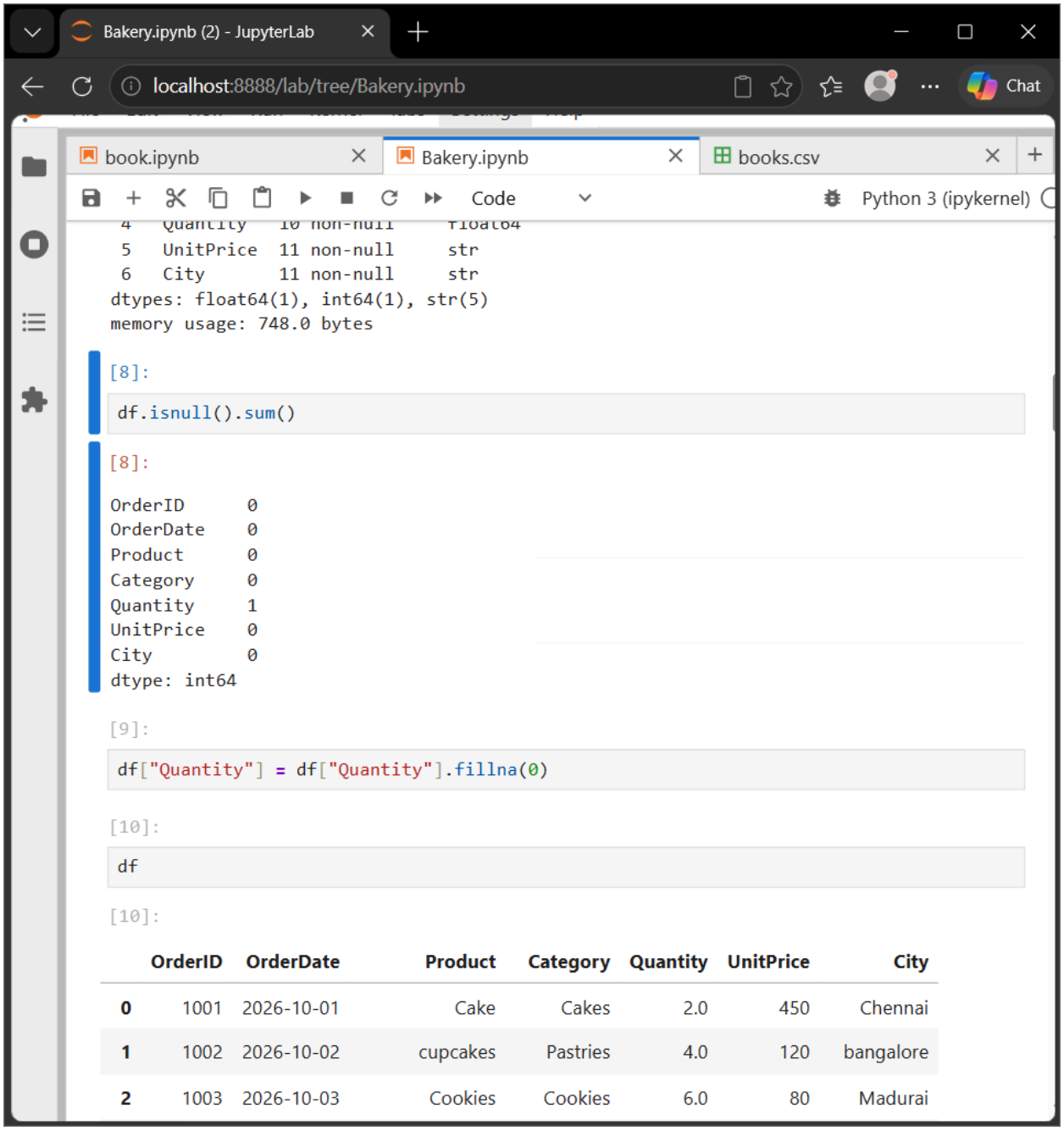
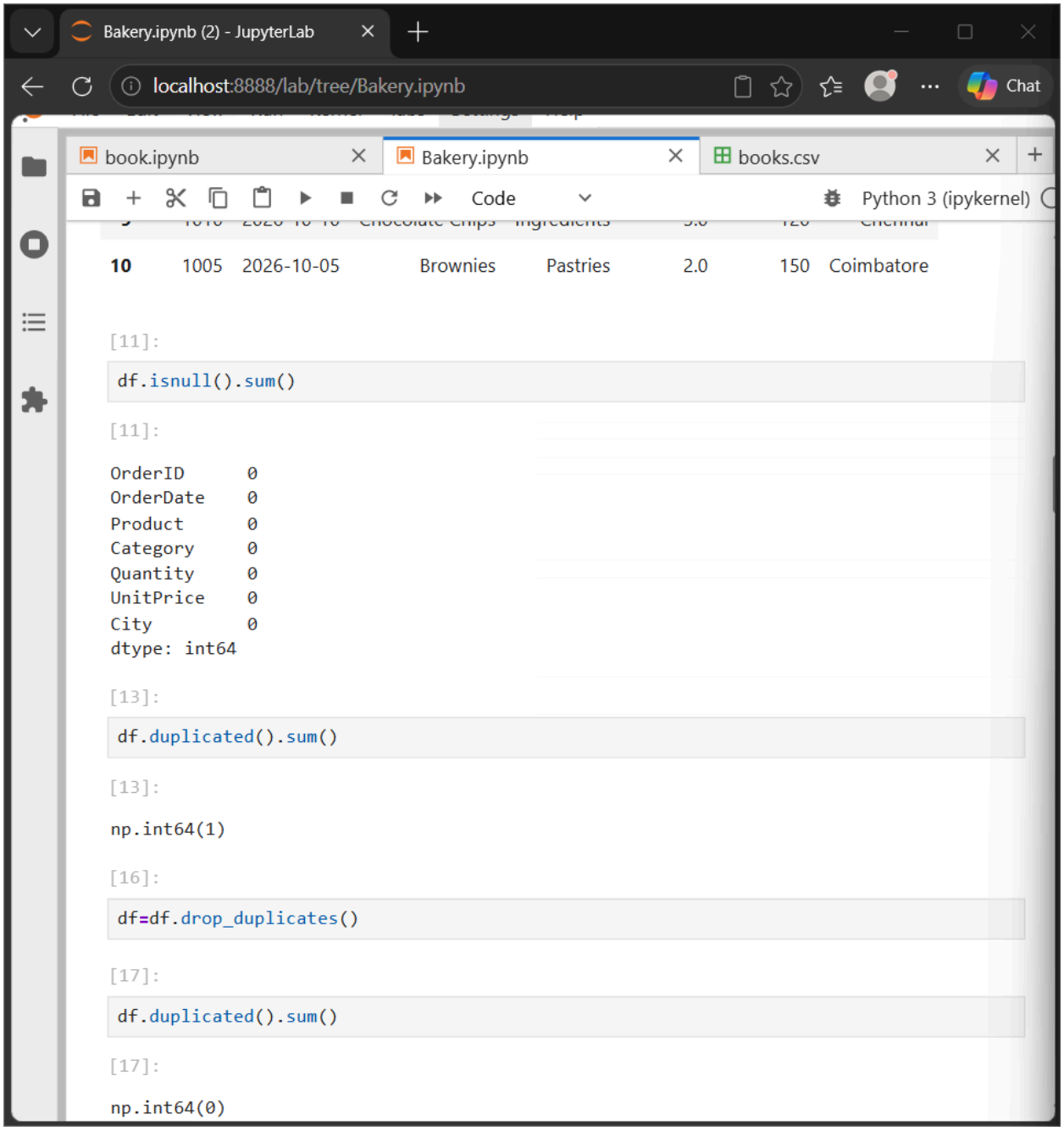
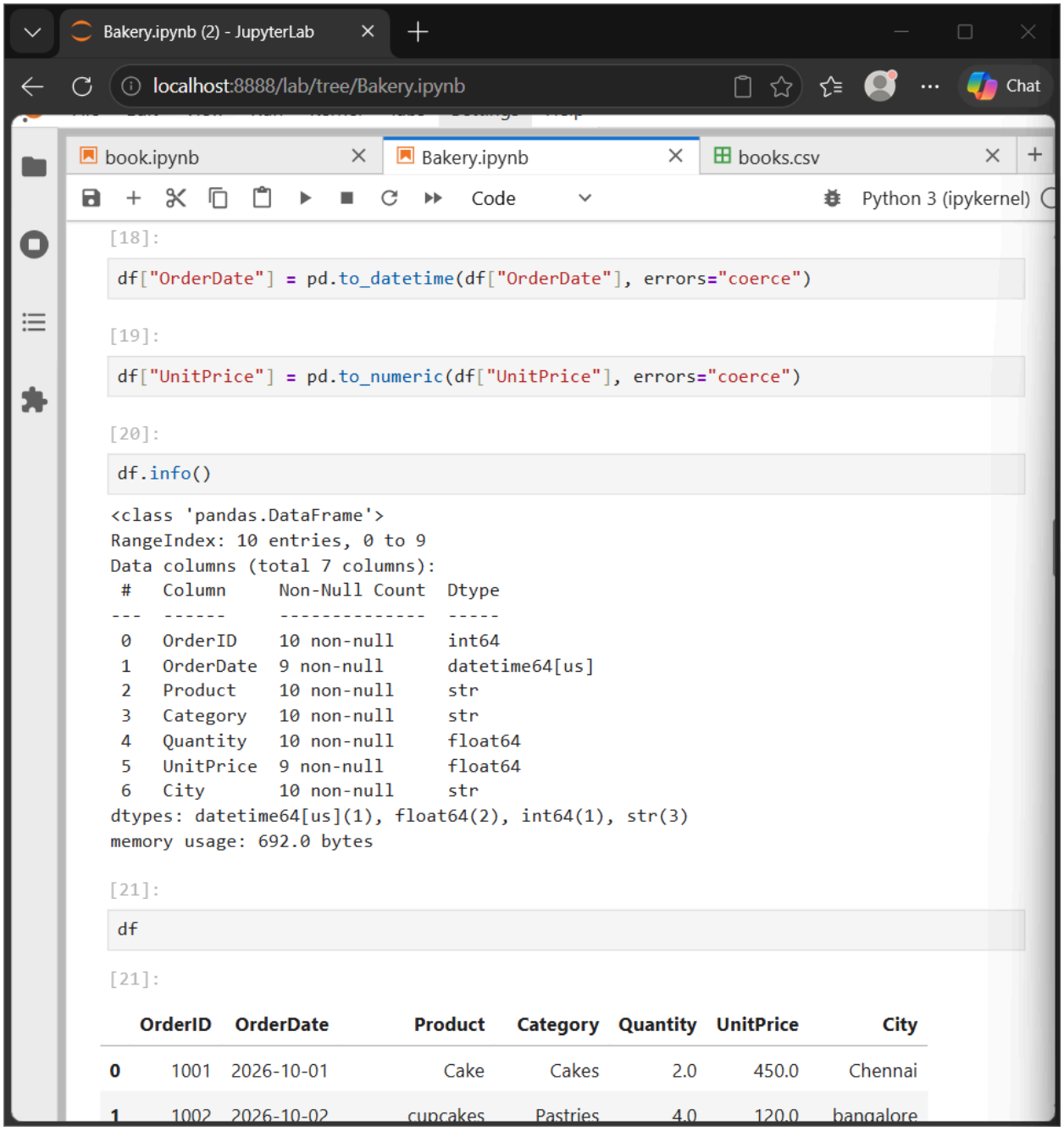

Bakery.ipynb (2) - JupyterLab
localhost:8888/lab/tree/Bakery.ipynb
Chat
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book.ipynb
Bakery.ipynb
books.csv
Code
Python 3 (ipykernel)
[2]:
import numpy as np
import pandas as pd
df = pd.read_csv("bakery_sales.csv")
print(df)
OrderID OrderDate Product Category Quantity UnitPrice \
0 1001 2026-10-01 Cake Cakes 2.0 450
1 1002 2026-10-02 cupcakes Pastries 4.0 120
2 1003 2026-10-03 Cookies Cookies 6.0 80
3 1004 not_a_date Bread Bread 3.0 60
4 1005 2026-10-05 Brownies Pastries 2.0 150
5 1006 2026-10-06 Muffins Pastries NaN 100
6 1007 2026-10-07 Croissants Pastries 4.0 180
7 1008 2026-10-08 Flour Ingredients 10.0 40
8 1009 2026-10-09 Baking Powder Ingredients 8.0 unknown
9 1010 2026-10-10 Chocolate Chips Ingredients 5.0 120
10 1005 2026-10-05 Brownies Pastries 2.0 150
City
0 Chennai
1 bangalore
2 Madurai
3 Chennai
4 Coimbatore
5 Bangalore
6 chennai
7 Madurai
8 Coimbatore
9 Chennai
10 Coimbatore
[3]:

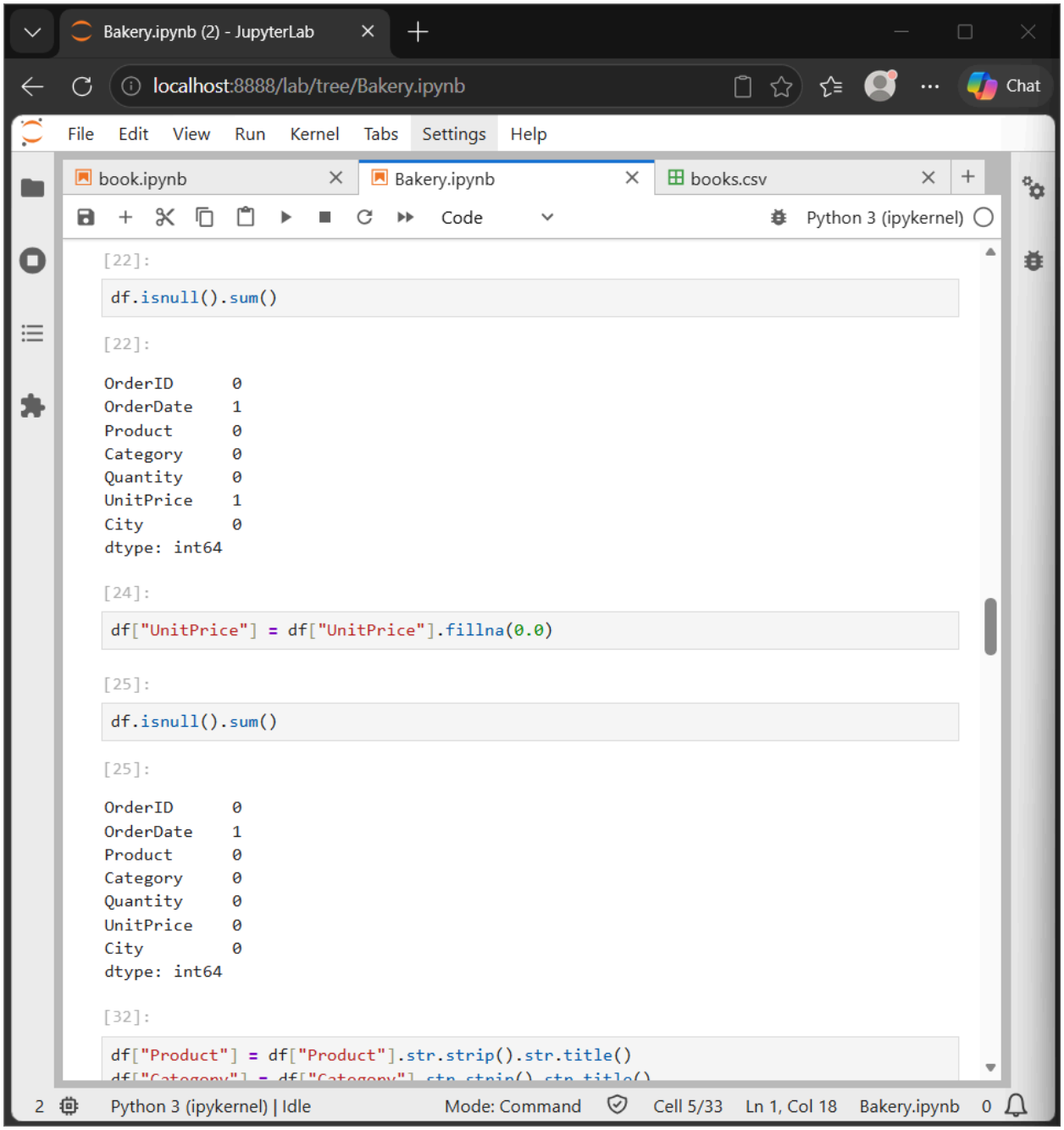


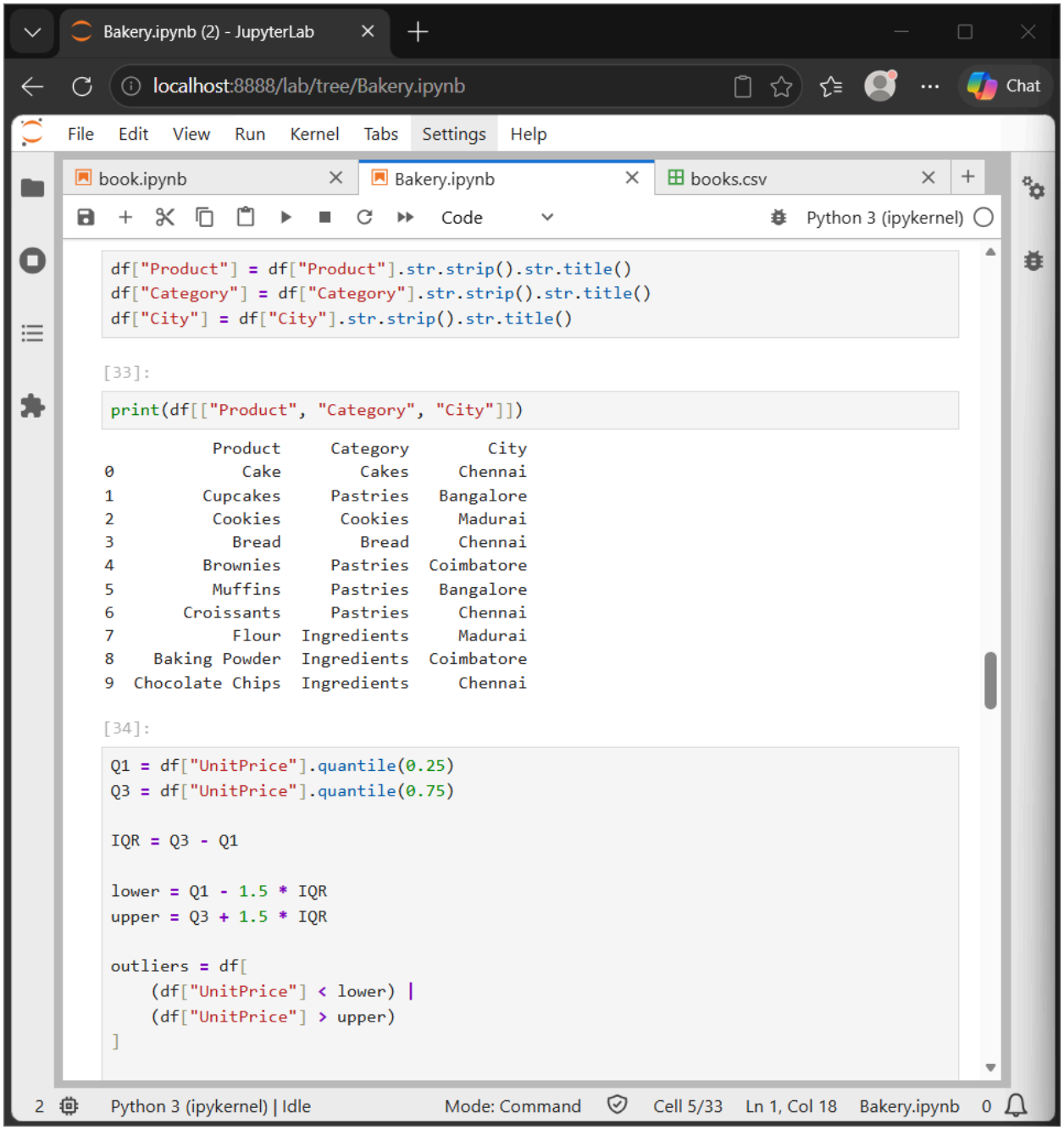
	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Cake	Cakes	2.0	450	Chennai
1	1002	2026-10-02	cupcakes	Pastries	4.0	120	bangalore
2	1003	2026-10-03	Cookies	Cookies	6.0	80	Madurai
3	1004	not_a_date	Bread	Bread	3.0	60	Chennai
4	1005	2026-10-05	Brownies	Pastries	2.0	150	Coimbatore


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Python 3 (ipykernel)
4 Quantity 10 non-null float64
5 UnitPrice 11 non-null str
6 City 11 non-null str
dtypes: float64(1), int64(1), str(5)
memory usage: 748.0 bytes
[8]:
df.isnull().sum()
[8]:
OrderID 0
OrderDate 0
Product 0
Category 0
Quantity 1
UnitPrice 0
City 0
dtype: int64
[9]:
df["Quantity"] = df["Quantity"].fillna(0)
[10]:
df
[10]:
OrderID OrderDate Product Category Quantity UnitPrice City
0 1001 2026-10-01 Cake Cakes 2.0 450 Chennai
1 1002 2026-10-02 cupcakes Pastries 4.0 120 bangalore
2 1003 2026-10-03 Cookies Cookies 6.0 80 Madurai


Bakery.ipynb (2) - JupyterLab
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Code
Python 3 (ipykernel)
10 1005 2026-10-05 Brownies Pastries 2.0 150 Coimbatore
[11]:
df.isnull().sum()
[11]:
OrderID 0
OrderDate 0
Product 0
Category 0
Quantity 0
UnitPrice 0
City 0
dtype: int64
[13]:
df.duplicated().sum()
[13]:
np.int64(1)
[16]:
df=df.drop_duplicates()
[17]:
df.duplicated().sum()
[17]:
np.int64(0)


Bakery.ipynb (2) - JupyterLab
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Code
Python 3 (ipykernel)
[18]:
df["OrderDate"] = pd.to_datetime(df["OrderDate"], errors="coerce")
[19]:
df["UnitPrice"] = pd.to_numeric(df["UnitPrice"], errors="coerce")
[20]:
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
Column Non-Null Count Dtype
--- ------ -------------- -----
0 OrderID 10 non-null int64
1 OrderDate 9 non-null datetime64[us]
2 Product 10 non-null str
3 Category 10 non-null str
4 Quantity 10 non-null float64
5 UnitPrice 9 non-null float64
6 City 10 non-null str
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 692.0 bytes
[21]:
df
[21]:
OrderID OrderDate Product Category Quantity UnitPrice City
0 1001 2026-10-01 Cake Cakes 2.0 450.0 Chennai
1 1002 2026-10-02 cupcakes Pastries 4.0 120.0 bangalore


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Code
Python 3 (ipykernel)
[22]:
df.isnull().sum()
[22]:
OrderID 0
OrderDate 1
Product 0
Category 0
Quantity 0
UnitPrice 1
City 0
dtype: int64
[24]:
df["UnitPrice"] = df["UnitPrice"].fillna(0.0)
[25]:
df.isnull().sum()
[25]:
OrderID 0
OrderDate 1
Product 0
Category 0
Quantity 0
UnitPrice 0
City 0
dtype: int64
[32]:
df["Product"] = df["Product"].str.strip().str.title()
2 Python 3 (ipykernel) | Idle Mode: Command Cell 5/33 Ln 1, Col 18 Bakery.ipynb 0


Bakery.ipynb (2) - JupyterLab
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Code
Python 3 (ipykernel)
df["Product"] = df["Product"].str.strip().str.title()
df["Category"] = df["Category"].str.strip().str.title()
df["City"] = df["City"].str.strip().str.title()
[33]:
print(df[["Product", "Category", "City"]])
Product Category City
0 Cake Cakes Chennai
1 Cupcakes Pastries Bangalore
2 Cookies Cookies Madurai
3 Bread Bread Chennai
4 Brownies Pastries Coimbatore
5 Muffins Pastries Bangalore
6 Croissants Pastries Chennai
7 Flour Ingredients Madurai
8 Baking Powder Ingredients Coimbatore
9 Chocolate Chips Ingredients Chennai
[34]:
Q1 = df["UnitPrice"].quantile(0.25)
Q3 = df["UnitPrice"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[
(df["UnitPrice"] < lower) |
(df["UnitPrice"] > upper)
]
2 Python 3 (ipykernel) | Idle Mode: Command Cell 5/33 Ln 1, Col 18 Bakery.ipynb 0

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Code | Python 3 (ipykernel)

```
print("Lower limit:", lower)
print("Upper limit:", upper)
print("Outliers:")
print(outliers[["Product", "UnitPrice"]])
```

```
Lower limit: -51.25
Upper limit: 258.75
Outliers:
  Product  UnitPrice
0    Cake      450.0
```

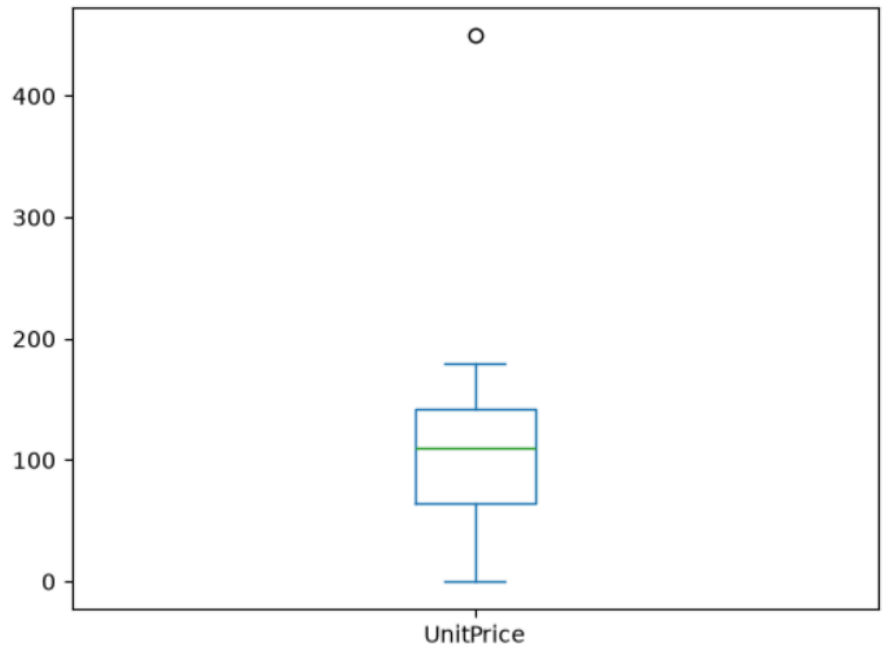
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book.ipynb | Bakery.ipynb | books.csv

Code | Python 3 (ipykernel)

[35]:

<Axes: >



[36]:

```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
print(df[["Product", "Quantity", "UnitPrice", "Revenue"]])
```

```
          Product  Quantity  UnitPrice  Revenue
0            Cake       2.0      450.0    900.0
1       Cupcakes        4.0      120.0    480.0
2         Cookies       6.0       80.0    480.0
```

[37]:

```
def price_category(price):
    if price < 100:
        return "Low"
    elif price <= 200:
        return "Medium"
    else:
        return "High"

df["PriceCategory"] = df["UnitPrice"].apply(price_category)

print(df[["Product", "UnitPrice", "PriceCategory"]])
```

```
           Product  UnitPrice PriceCategory
0             Cake      450.0          High
1         Cupcakes      120.0        Medium
2          Cookies       80.0           Low
3            Bread       60.0           Low
4         Brownies      150.0        Medium
5          Muffins      100.0        Medium
6       Croissants      180.0        Medium
7            Flour       40.0           Low
8    Baking Powder        0.0           Low
9  Chocolate Chips      120.0        Medium
```

[38]:

```
df.to_csv("bakery_sales_cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```